LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Motion

Chapter / Topic: Motion (topic-based)

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SECTION A

Q1. The SI unit of acceleration is: **[1]**

Answer: (d) m/s^2

Q2. The slope of a velocity–time graph gives: **[1]**

Answer: (c) acceleration

Q3. A body moving along a circular path at constant speed has: **[1]**

Answer: (c) changing velocity (its direction keeps changing)

Q4. A car goes from rest to 20 m/s in 4 s. Its acceleration is: **[1]**

Answer: (b) 5 m/s^2

Q5. Which of these can be zero even when a body has moved? **[1]**

Answer: (d) Displacement

Q6. **Assertion (A):** A body moving with uniform velocity has zero acceleration. **Reason (R):** Acceleration is the rate of change of velocity. **[1]**

Answer: (a)

SECTION B

Q7. Give two differences between distance and displacement. **[2]**

Answer: Distance is the total length of the path travelled, while displacement is the shortest distance between the initial and final positions. Distance has only magnitude (scalar), while displacement has magnitude and direction (vector). Distance is never negative or zero for a moving body, while displacement can be zero.

- Any two differences: 1 + 1

OR, answer: Speed is distance per unit time, while velocity is displacement per unit time. Speed has only magnitude, while velocity also has direction. Velocity can be zero for a moving body (e.g. after a round trip); its average speed cannot.

- Any two differences: 1 + 1

Q8. A cyclist completes one round of a circular track of radius 21 m in 44 s. Find the cyclist's speed. (Take $\pi = 22/7$.) **[2]**

Answer: Distance = $2\pi r = 2 \times (22/7) \times 21 = 132$ m. Speed = $132/44 = 3$ m/s.

- Circumference: 1
- Speed: 1

SECTION C

Q9. A train speeds up uniformly from 10 m/s to 30 m/s in 10 s. **[3]**

(i) $a = (30 – 10)/10 = 2\ m/s^2$ [1]

(ii) $s = ut + \frac{1}{2}at^2 = 10 \times 10 + \frac{1}{2} \times 2 \times 100 = 200$ m [1]

(iii) $v = 30 + 2 \times 5 = 40$ m/s [1]

Q10. Derive the equation $v = u + at$ using a velocity–time graph. **[3]**

Answer: Draw a v–t graph: velocity u at time 0 (point A) rising uniformly to v at time t (point B). The acceleration equals the slope: a = (change in velocity)/time = $(v – u)/t$. Rearranging gives $v = u + at$.

- Graph described/drawn: 1
- Slope = acceleration: 1
- Rearrangement: 1

OR, answer: u = 72 km/h = 20 m/s, v = 0, t = 4 s. a = (0 – 20)/4 = –5 m/s^2 (retardation 5 m/s^2). s = (v^2 – u^2)/2a = (0 – 400)/(–10) = 40 m.

- Unit conversion: 1
- Retardation: 1
- Distance: 1

SECTION D

Q11. Answer the following: **[5]**

(i) The distance travelled equals the area under the v–t graph. The area is a rectangle (height u, width t), giving ut, plus a triangle (base t, height v – u = at), giving ½ × t × at = ½at^2. So s = ut + ½at^2. [3]

- Distance = area under graph: 1
- Rectangle + triangle areas: 1
- Final equation: 1

(ii) u = 0, so s = ½at^2: 9 = ½ × a × 9, which gives a = 2 m/s^2. [2]

- Formula: 1
- a = 2 m/s^2: 1

SECTION E

Q12. A student recorded the distance travelled by a scooter on a straight road: at t = 0, 2, 4 and 6 s, the distance from the start was 0, 10, 20 and 30 m. **[4]**

(i) Uniform motion: equal distances in equal intervals of time. [1]

(ii) Speed = 10 m ÷ 2 s = 5 m/s. [1]

(iii) Distance = 5 × 10 = 50 m. The graph is a straight line through the origin. [2]

- 50 m: 1
- Straight line through origin: 1